BALANCEBITE

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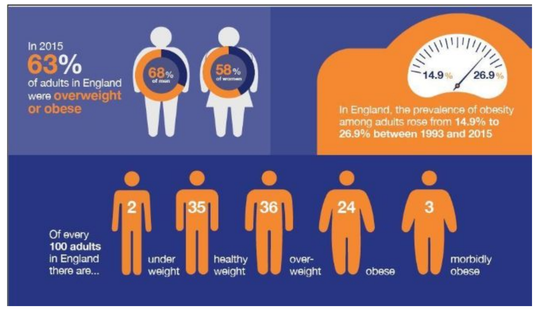
ABSTRACT

Worldwide obesity preys on 50 percent of Pakistani population who carry excessive body weight. The increased rate of obesity creates additional threats to Pakistan's population for the development of chronic illnesses including heart disease and diabetes. The fast-expanding mobile phone market in Pakistan with its 192 million active users creates an optimal environment to fight obesity using creative digital solutions.

BalanceBite represents an AI-powered mobile application which monitors dietary intake and exercise habits of users using sensors and machine learning algorithms to process images of food and evaluate caloric values while measuring bodily movements through sensor analysis for caloric expenditure. The vision transformer model has achieved 90% accuracy. The project selects South-Asian cuisine as its main focus because it wants to connect with its local cultural background.

INTRODUCTION

One of the 21st century's most urgent global public health issues is obesity, which is becoming more widely acknowledged. Obesity is characterized by abnormal or excessive fat buildup that poses a health concern, according to the World Health Organization (WHO). With at least 2.8 million deaths annually attributed to overweight or obesity, obesity has reached epidemic proportions worldwide [1]. Obesity, often thought to be a problem in high-income nations, is now common in low- and middle- income nations as well. Global obesity rates have almost tripled since 1975, according to WHO data [1]. Increased consumption of high-calorie and low-nutrient foods, sedentary lifestyles, urbanization, and technology improvements that encourage physical inactivity are some of the reasons contributing to the worldwide obesity epidemic's acceleration in recent decades.



LITERATURE REVIEW

It is possible that AI recognition of items from a photo taken by a user and an evaluation of their nutritional value could accompany dietary recommendations. In parts of South Asian nations, where native meals might not appropriately be represented or may be omitted completely in generalized eating plans, this technique is quite beneficial.

Feature	CalorieMama AI [11]	HealthifyMe [10]	Lose It! [12]	Foodvisor [9]	BalanceBite
Sensors Used	Camera (for image recognition), Accelerometer (optional for activity tracking)	Accelerometer, GPS	Barcode Scanner, Snap It, Accelerometer, GPS	Barcode Scanner, Snap It, Accelerometer, GPS	Camera, Accelerometer, Gyroscope, Ambient Light Sensor
Classifiers Used	Convolutional Neural Networks (CNN) for image classification	Deep Learning Models	Barcode scanner classifier, food item matching from database	Deep Learning Models	Convolutional Neural Networks (CNN) for food image classification
What does it identify?	Identifies food types, portion sizes, and nutrients from images using AI- based food recognition	Manually identifies food items from database based on user selection	Identifies packaged foods using barcodes, matching with database entries	Identifies food types, portion sizes, and nutrients from images using AI	Identifies food types, portion sizes, and nutrients from images using AI; user can select quantity or manually enter it
Sensors for Activity Tracking	Syncs with external sensors like Apple Watch, Google Fit for step counting, sleep tracking	Accelerometer, GPS Fit for tracking steps and activities	Syncs with Apple Health, Google Fit for step counting, physical activity	Syncs with Apple Health, Google Fit for activity tracking	Uses only Phone sensors: Accelerometer and Gyroscope (Angular) for activity tracking

Another essential component of the digital health tools that support weight management is the activity recognition. The smartphone sensors such as accelerometers and gyroscopes can be used to capture and monitor features like walking, running, and cycling.

REQUIREMENT SPECIFICATIONS

The suggested BalanceBite application solves current system problems by applying Vision Transformer AI technology with machine learning to deliver better performance. The Flutter application functions across multiple platforms to offer these features to users.

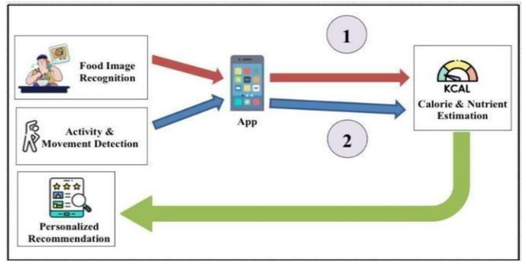
Users can also manually log their food by selecting them from a database. eliminating the need for the help of another person.

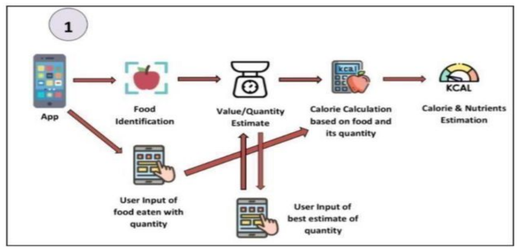
The program takes advantage of the accelerometers and gyroscopes that are installed in smartphones with real time measuring, logging of physical activity.

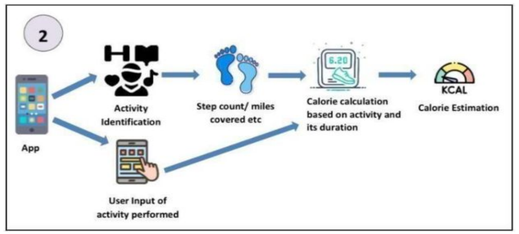
The program incorporates nice progress displays such graphs and charts; to increase motivation.

DESIGN AND METHODOLOGY

For food classification, we chose the Vision Transformer (ViT) model because it excels at understanding the relationships across different parts of an image. Unlike traditional CNNs, ViT breaks images into patches and processes them in sequence, which makes it better at recognizing the detailed textures and structures in food images.







SYSTEM IMPLEMENTATION

The BalanceBite application designers have created a system architecture that serves both users well and works as expected. Different Android users can use the app because its development with Flutter made cross-platform mobile apps easier to build.

The Vision Transformer (ViT) model serves as the primary component to identify dishes within BalanceBite application. Our system receives training from a special dataset made of 300 South Asian food items and their essential photos plus nutritional stats.

SYSTEM EVALUATION

Our assessment ensures BalanceBite works properly and meets its set requirements. Several basic and advanced tests took place in this phase including unit testing, integration testing, model testing and user acceptance testing (UAT). Our tests checked how the application works both in its user interface and system.

When users entered meal information, the program showed excellent responsiveness, especially when AI-based food recognition using image capture was used.

The ViT model correctly identified real dishes during more than 90% of the prediction attempts.

CONCLUSION

By combining culturally relevant meal planning using artificial intelligence, namely the Vision Transformer (ViT) model, the app's advocacy for the unique dietary and health concerns of users in the region. The users can make proper decisions of their diet and level of fitness. Moreover its ability of tracking physical activity and personalized health advice, the vision transformer model has achieved 90% accuracy.

REFERENCES

[1]
WHO, "World Health Organization," 7 May 2025. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight. [Accessed 8 May 2025].
[2]
"International Journal of Obesity," 17 November 2008. [Online]. Available: https://www.nature.com/articles/ijo2008186. [Accessed 8 May 2025].
[3]
P. H. England, "Health matters: obesity and the food environment," 21 December 2017. [Online]. Available: https://www.gov.uk/government/publications/health-matters-obesity-and-the-food-environment/health-matters-obesity-and-the-food-environment--2. [Accessed 8 May 2025].
[4]
"ScienceDirect," 6 May-June 2014. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0033062014000279?via%3Dihub. [Accessed 8 May 2025].

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